



Virtual Machines

Get operating systems to run on your operating system!

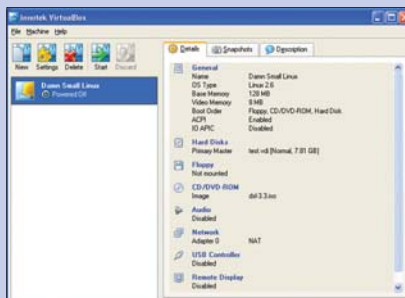
Rossi Fernandes

Some of us get into that weird state of mind where we want to just install a software to play around with it a bit. When you're thinking of it, there's that bit of hesitation when it comes to Linux, thanks to the general fear of partitioning and formatting of drives. What we're about to do is, get a Linux distribution running without a single boot and without a single drive partitioned or formatted. Impossible? We'll show you how!

VMWare has made a name for itself in the virtualisation software business. Of late, an open source virtualisation program called VirtualBox has been making waves. The way virtualisation works in VirtualBox, like in VMWare software, is that virtual components are created. First, download VirtualBox from www.virtualbox.org and install it.

Creating A New Virtual Machine

Remember, you can install multiple operating systems on VirtualBox. VirtualBox won't just let you install an



Damn Small Linux, ready to run off a Virtual Machine

operating system, it'll also allow you to run Linux LiveCDs without having to reboot. Either way, you need to create a virtual machine.

Click on the New button on the toolbar. Click Next and give a name to the operating system and the type (Windows XP, Linux, etc.) you plan on using. Click Next. Here, you choose how much RAM you want dedicated to the virtual machine. In the case of large OSes, keep a fairly large amount of memory.

Next, you choose the size of your disk drive; in the case of a virtual machine, you choose the size of the image. Click New to create a new image. Click Next

and you can choose from a fixed or an expanding image. A fixed image will allocate space in advance—so if you have limited hard drive space to offer the new OS, choose this. In the case of an expanding image, as the demand for space increases, the image file will increase in size, so you need to keep tabs on it to avoid running out of space on your hard disk. Click Next and enter a name for image file itself, and the size for it. Follow the next two steps to complete the procedure.

Using An Image To Install

Every time you download a Linux ISO, you have to burn a DVD or CD and install it. There's usually no other way to go about it. VirtualBox allows you to install off the ISO using emulation—the same way Alcohol 120% works.

Assuming you have the disk image parameters set up, click the virtual machine and click the Start button. This is similar to powering up a real computer by pressing the power button. A window will appear asking you to select the source to be used to install the OS on our virtual machine. Click Next, and choose whether you want to use the CD / DVD drive or a floppy or even an image, which is what we are using. To choose an ISO image, click the Image file radio button under Media source and point to the location of the image on your drive. The virtual machine will then boot using the data from the disc image. You can simply run the LiveCD (if you're using a Live CD image, of course) or install the entire OS.

Saving And Restoring States

Save a machine state so you can continue from where you left off. Hibernation in Windows is a feature that stores the current state of the operating system, which can be restored at a later time. The Snapshots feature in VirtualBox does the same thing. To save a session, close the



Save a machine state so you can continue from where you left off

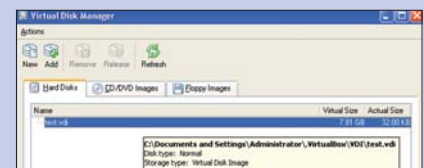
window for the virtual machine in use. Instead of choosing "Power off the machine", click "Save the machine state". The next time you power on the virtual machine, it will resume from the state you left off at.

Adding Extra Hard Drives To A Virtual Machine

In virtual computing, adding virtual hard drives isn't going to cost you anything. If you run out of space on the main image you created, you can increase your virtual disk space by adding additional images.

Click on the virtual machine and then on the Settings icon on the toolbar. Click on Hard Drives and choose Primary Slave or Secondary Slave. You can choose from one of the created images, or click on the Select icon to the right. Here, click New to create a new image, and follow the same steps you used in the creation of the virtual machine.

You can, at any time, change storage space resources by using the Virtual Disk Manager. Select File > Virtual Disk Manager. Here you can add, remove, and release hard disk, CD / DVD images, and floppy images.



Handling of images can be done by using the Virtual Disk Manager

Running Full-screen Mode

Running an OS in a window mode just doesn't seem right. Set up the resolution in the operating system properly and you can switch to full-screen mode by pressing [Right Ctrl] + [F]. To switch back to window mode, press [Right Ctrl] + [F] again.

For this demonstration, we used Damn Small Linux, but you can use always full-fledged distributions like Fedora (from this month's DVD) or Ubuntu. You can even run another instance of Windows on VirtualBox, or run a Windows virtual machine on Linux! There are plenty of things you can use VirtualBox for without having to worry about messing around with your current OS. ☑

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